

Avinash Bhojanapalli

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Summary

Software & Machine Learning Engineer with a Master's in Computer Science specializing in Data Science. Experienced in bridging the gap between data exploration and production deployment. Proficient in developing scalable backend services (**Go, Python**) and optimizing computationally heavy inference pipelines (**Rust, PyTorch**). Demonstrated ability to build robust REST APIs, containerize applications with Docker/Kubernetes, and automate workflows to reduce latency and improve system reliability. Passionate about applying core software engineering principles to modern AI challenges.

EDUCATION

Rochester Institute of Technology

MS Computer Science (Data Science) (GPA: 3.9/4.0)

Aug 2024 - Apr 2026

Rochester, NY

IIIT Sri City

BTech Computer Science and Engineering

Dec 2020 - May 2024

India

SKILLS

- **Languages:** Python, C++, Go, Rust, SQL, Bash
- **Backend & Systems:** RESTful APIs, Microservices, FastAPI, Linux, Git
- **MLOps & Cloud:** Docker, Kubernetes, AWS (SageMaker, Lambda, ECR), CI/CD (GitHub Actions), MLflow, GCP (Vertex AI), Grafana
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Pandas, HuggingFace, Langchain, ONNX
- **Data & Databases:** PostgreSQL, MongoDB, Redis, ChromaDB (Vector DB), Apache Kafka, Neo4j, SQLite

EXPERIENCE

dCortex | *Founding Machine Learning Engineer*

May 2026 - Aug 2026

- Built a **live 72-hour flight-crew risk model** that scores disruption and legality exposure so controllers can act before coverage gaps cascade.
- Built a **counterfactual analysis** platform for the crew recommendation service that finds the ranking changes that would have placed a controller's chosen crew first.
- Built a **chat agent** that writes and runs **SQL** on the fly so controllers can fetch crew identity, reserve status, and legality details in plain language.

Bonsai Lab RIT | *Machine Learning Engineer*

Aug 2025 - Apr 2026

- Architected a modular **Self-Supervised Learning (SSL)** pipeline using **PyTorch**, enabling rapid testing and comparison of model representations and reducing experiment iteration time by **60%**.
- Optimized data loading pipelines with pre-fetching and caching, which increased GPU utilization and reduced training time.
- Implemented automated experiment tracking and artifact versioning, ensuring reproducibility across distributed training runs and standardizing metric reporting for team-wide collaboration.

Incribo | *MLOps Engineer Intern*

Jan 2024 - Jul 2024

- Designed and deployed a fault-tolerant microservices architecture for multi-agent AI systems on Kubernetes and Docker, using RabbitMQ for robust asynchronous messaging, which improved system reliability and enabled seamless scaling of AI agents
- Optimized **Large Language Model (LLM)** serving pipelines by implementing **gradient checkpointing** and dynamic batching, resulting in a **15% reduction in GPU memory usage** and improved inference latency.
- Developed a **Go-based** CLI tool to automate container orchestration and deployment workflows, reducing deployment friction and ensuring consistent environments across development and production.
- Integrated **Synthetic Data Generation** workflows using neuroevolution algorithms (CPPN-NEAT), scaling data generation rates by **15%** to support robust model training and validation.

KEY PROJECTS

Crosstem: High-Performance NLP Library | Rust, Python, PyO3, CI/CD | PyPI

- Engineered a multilingual NLP stemming library with a Rust-accelerated core (PyO3) and Python API, delivering 2.7-2.9x speedup over the Python fallback and 15x higher throughput than Porter on benchmark workloads.
- Built a graph-based morphological stemming pipeline (BFS + productivity-threshold scoring) that resolves overstemming and correctly maps across part-of-speech boundaries.
- Shipped and maintained the library on PyPI with parity-tested Rust/Python backends, production packaging, and end-to-end documentation for installation, API usage, and performance validation.

Ledgerline: Vectorless RAG Intelligence Engine | Go, Python, Kafka, Redis | Github

- Engineered an event-driven, vectorless RAG microservices stack, replacing traditional chunking with hierarchical tree navigation to eliminate context loss across complex 200+ page financial filings.
- Architected a high-throughput API Gateway in Go (Fiber) featuring WebSocket streaming for real-time AI reasoning, integrating Redis caching to hot-load document trees and maintain query latency strictly under 3 seconds.
- Orchestrated asynchronous Kafka pipelines for reliable service handoff, integrating an automated LLM-as-a-judge observability system backed by PostgreSQL to continuously evaluate response telemetry and rigorously prevent hallucinations.

Topological Arbitrage Engine: Market-Neutral Quant Strategy | Python, Pandas, SciPy, Flask | Github

- Developed a statistical arbitrage trading engine applying Graph Laplacian diffusion to CAPM residuals, isolating idiosyncratic alpha to exploit structural mean-reversion across a dynamic 80-stock universe.
- Architected a rigorous walk-forward validation pipeline that proved the raw topological signal thrives during market volatility, achieving a 42.5% annualized return and a 1.99 Sharpe ratio on strictly out-of-sample 2025-2026 data.

PUBLICATIONS

- A Bhojanapalli, et al. A CALIPSO Observation Based 3-Dimensional Tropospheric Aerosol Classification Model Over the Indian City Delhi. In 2023 IEEE International Geoscience and Remote Sensing Symposium, pp. 305-308. IEEE. 16-21 July 2023, Pasadena, CA, USA. DOI:10.1109/IGARSS52108.2023.10282609
- Z Lyu, et al. Evolving RNNs for Stock Forecasting: A Low Parameter Efficient Alternative to Transformers. In International Conference on the Applications of Evolutionary Computation (Part of EvoStar), pp. 36-54. Springer, Cham. 17 April 2025, Trieste, Italy. DOI:10.1007/978-3-031-90065-5_3